

DAY 2: INTEGRATION - NOTES

ANTI-DERIVATES

Goal

Achieve fluency in ALL integration techniques (power rule, u-substitution, integration by parts, trigonometric substitution, partial fractions) and apply them to physical problems including work, energy, center of mass, moment of inertia, and volumes of irregular shapes.

1 THE FUNDAMENTAL THEOREM OF CALCULUS

1.1 The Big Idea

Integration is the inverse operation of differentiation. While derivatives tell us how fast something changes, integrals tell us how much has accumulated.

1.2 Fundamental Theorem of Calculus Part 1

If f is continuous on $[a, b]$ and $F(x) = \int_a^x f(t)dt$, then:

$$\boxed{F'(x) = f(x)}$$

1.3 Fundamental Theorem of Calculus Part 2

If F is an antiderivative of f (i.e., $F' = f$), then:

$$\boxed{\int_a^b f(x)dx = F(b) - F(a)}$$

2 BASIC INTEGRATION RULES

2.1 Power Rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

2.2 Special Cases

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\int e^x dx = e^x + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C$$

2.3 Trigonometric Integrals

$$\int \sin x \, dx = -\cos x + C$$

$$\int \cos x \, dx = \sin x + C$$

$$\int \sec^2 x \, dx = \tan x + C$$

$$\int \csc^2 x \, dx = -\cot x + C$$

$$\int \sec x \tan x \, dx = \sec x + C$$

$$\int \csc x \cot x \, dx = -\csc x + C$$

$$\int \tan x \, dx = -\ln |\cos x| + C = \ln |\sec x| + C$$

$$\int \cot x \, dx = \ln |\sin x| + C$$

2.4 Inverse Trigonometric Forms

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin\left(\frac{x}{a}\right) + C$$

$$\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$$

$$\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a} \operatorname{arcsec}\left(\frac{|x|}{a}\right) + C$$

3 INTEGRATION TECHNIQUES

3.1 U-Substitution (Reverse Chain Rule)

If $u = g(x)$, then $du = g'(x)dx$ and:

$$\int f(g(x))g'(x)dx = \int f(u)du$$

For definite integrals:

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du$$

Examples of when to use:

- Integrals containing $f(g(x)) \cdot g'(x)$
- $\int \sin(ax + b)dx$ (let $u = ax + b$)
- $\int xe^{x^2}dx$ (let $u = x^2$)
- $\int \frac{2x}{x^2+1}dx$ (let $u = x^2 + 1$)

3.2 Integration by Parts (Reverse Product Rule)

$$\int u dv = uv - \int v du$$

Choice of u and dv : Use LIATE rule (choose u in this order):

- **L:** Logarithmic functions ($\ln x$)
- **I:** Inverse trigonometric functions ($\arcsin x$, $\arctan x$)
- **A:** Algebraic functions (x^n)
- **T:** Trigonometric functions ($\sin x$, $\cos x$)
- **E:** Exponential functions (e^x)

Examples:

- $\int xe^x dx$ (let $u = x$, $dv = e^x dx$)

- $\int x \sin x dx$ (let $u = x$, $dv = \sin x dx$)
- $\int \ln x dx$ (let $u = \ln x$, $dv = dx$)

3.3 Trigonometric Substitution

Used when integrand contains $\sqrt{a^2 \pm x^2}$ or $\sqrt{x^2 \pm a^2}$.

Expression	Substitution	Identity
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$, $dx = a \cos \theta d\theta$	$1 - \sin^2 \theta = \cos^2 \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$, $dx = a \sec^2 \theta d\theta$	$1 + \tan^2 \theta = \sec^2 \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$, $dx = a \sec \theta \tan \theta d\theta$	$\sec^2 \theta - 1 = \tan^2 \theta$

Examples:

- $\int \sqrt{4 - x^2} dx$ (let $x = 2 \sin \theta$)
- $\int \frac{dx}{\sqrt{x^2 + 9}}$ (let $x = 3 \tan \theta$)
- $\int \frac{dx}{\sqrt{x^2 - 16}}$ (let $x = 4 \sec \theta$)

3.4 Partial Fractions

Used for rational functions $\frac{P(x)}{Q(x)}$ where degree of P $\leq$ degree of Q .

Procedure:

1. Factor $Q(x)$ into linear and irreducible quadratic factors.
2. For each linear factor $(ax + b)^n$:

$$\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \cdots + \frac{A_n}{(ax + b)^n}$$

3. For each quadratic factor $(ax^2 + bx + c)^m$:

$$\frac{B_1x + C_1}{ax^2 + bx + c} + \cdots + \frac{B_mx + C_m}{(ax^2 + bx + c)^m}$$

4. Solve for constants by clearing denominators and equating coefficients.
5. Integrate term by term.

Examples:

- $\int \frac{1}{x^2 - 4} dx = \int \left(\frac{1/4}{x - 2} - \frac{1/4}{x + 2} \right) dx$
- $\int \frac{2x + 3}{x^2 + 3x + 2} dx = \int \left(\frac{1}{x + 1} + \frac{1}{x + 2} \right) dx$

3.5 Improper Integrals

$$\int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx$$

$$\int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx$$

$$\int_{-\infty}^\infty f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^\infty f(x) dx$$

The integral **converges** if the limit exists and is finite; otherwise it **diverges**.

Examples:

- $\int_1^\infty \frac{1}{x^2} dx$ converges to 1
- $\int_1^\infty \frac{1}{x} dx$ diverges

4 PHYSICS APPLICATIONS OF INTEGRATION

4.1 Kinematics

From acceleration to velocity:

$$v(t) = \int a(t) dt + v_0$$

From velocity to position:

$$x(t) = \int v(t) dt + x_0$$

Displacement (net change in position):

$$\Delta x = \int_{t_1}^{t_2} v(t) dt$$

Total distance traveled:

$$\text{Distance} = \int_{t_1}^{t_2} |v(t)| dt$$

Example: If $a(t) = 6t \text{ m/s}^2$, $v(0) = 2 \text{ m/s}$, then:

$$v(t) = \int 6t dt = 3t^2 + 2 \text{ m/s}$$

4.2 Work

Work done by a variable force $F(x)$:

$$W = \int_{x_1}^{x_2} F(x) dx$$

Spring (Hooke's Law): $F(x) = kx$

$$W = \int_0^x kx dx = \frac{1}{2}kx^2$$

Work to lift a chain/rope: For a chain of length L with linear density λ kg/m, lifting from one end:

$$W = \int_0^L \lambda g x \, dx = \frac{1}{2} \lambda g L^2$$

Work to pump liquid out of a tank: For a tank of height H , cross-sectional area $A(y)$, pumping to height h_0 :

$$W = \int \rho g \cdot (\text{distance lifted}) \cdot dV$$

4.3 Mass from Density

For a rod along the x-axis with linear density $\lambda(x)$:

$$M = \int_a^b \lambda(x) dx$$

For a 2D lamina with area density $\sigma(x, y)$:

$$M = \iint_R \sigma(x, y) dA$$

For a 3D object with volume density $\rho(x, y, z)$:

$$M = \iiint_D \rho(x, y, z) dV$$

4.4 Center of Mass

For a rod with linear density $\lambda(x)$:

$$x_{cm} = \frac{1}{M} \int_a^b x \lambda(x) dx$$

For a 2D lamina with density $\sigma(x, y)$:

$$x_{cm} = \frac{1}{M} \iint_R x \sigma(x, y) dA, \quad y_{cm} = \frac{1}{M} \iint_R y \sigma(x, y) dA$$

For a 3D object with density $\rho(x, y, z)$:

$$x_{cm} = \frac{1}{M} \iiint_D x \rho(x, y, z) dV$$

$$y_{cm} = \frac{1}{M} \iiint_D y \rho(x, y, z) dV$$

$$z_{cm} = \frac{1}{M} \iiint_D z \rho(x, y, z) dV$$

4.5 Moment of Inertia

The moment of inertia measures an object's resistance to rotational acceleration about an axis.

$$I = \int r^2 dm$$

where r is the perpendicular distance from the axis of rotation.

For a rod with linear density $\lambda(x)$ about an axis at one end:

$$I = \int_0^L x^2 \lambda(x) dx$$

For a 2D lamina about an axis perpendicular to its plane:

$$I = \iint_R r^2 \sigma(x, y) dA$$

For a 3D object about the z-axis:

$$I_z = \iiint_D (x^2 + y^2) \rho(x, y, z) dV$$

About the x-axis:

$$I_x = \iiint_D (y^2 + z^2) \rho(x, y, z) dV$$

About the y-axis:

$$I_y = \iiint_D (x^2 + z^2) \rho(x, y, z) dV$$

Common moments of inertia (uniform density):

- Thin hoop about center: $I = MR^2$
- Solid cylinder about axis: $I = \frac{1}{2}MR^2$
- Solid sphere about diameter: $I = \frac{2}{5}MR^2$
- Thin rod about center: $I = \frac{1}{12}ML^2$
- Thin rod about end: $I = \frac{1}{3}ML^2$

4.6 Parallel Axis Theorem

If we know the moment of inertia about an axis through the center of mass:

$$I = I_{cm} + Md^2$$

where d is the perpendicular distance from the center of mass to the new parallel axis.

Example: For a rod of length L , $I_{cm} = \frac{1}{12}ML^2$. About one end ($d = L/2$):

$$I = \frac{1}{12}ML^2 + M\left(\frac{L}{2}\right)^2 = \frac{1}{12}ML^2 + \frac{1}{4}ML^2 = \frac{1}{3}ML^2$$

4.7 Hydrostatic Force

Force on a submerged vertical surface:

$$F = \int \rho g \cdot (\text{depth}) \cdot (\text{width}) dy$$

where ρ is fluid density, g is gravity, depth is distance from surface.

Example: For a rectangular plate of width w extending from depth a to b :

$$F = \int_a^b \rho g y \cdot w dy = \frac{1}{2} \rho g w (b^2 - a^2)$$

4.8 Average Value

Average value of $f(x)$ on $[a, b]$:

$$f_{avg} = \frac{1}{b-a} \int_a^b f(x) dx$$

Example: Average velocity over time interval:

$$v_{avg} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} v(t) dt$$

4.9 Root Mean Square (RMS)

$$f_{rms} = \sqrt{\frac{1}{b-a} \int_a^b [f(x)]^2 dx}$$

Important in AC circuits: For $V(t) = V_0 \sin(\omega t)$ over one period $T = 2\pi/\omega$:

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T V_0^2 \sin^2(\omega t) dt} = \frac{V_0}{\sqrt{2}}$$

5 VOLUME OF IRREGULAR SHAPES

5.1 General Approach: Slice, Approximate, Integrate

The fundamental strategy for finding volumes of irregular shapes:

1. **Slice** the object into thin pieces
2. **Approximate** the volume of each slice
3. **Integrate** to sum all slices

5.2 Volume by Cross-Sections

If we know the area $A(x)$ of each cross-section perpendicular to the x-axis:

$$V = \int_a^b A(x) dx$$

Example: A pyramid with square base of side s and height h . At height y from the apex, cross-section side length $= \frac{s}{h}y$, area $A(y) = \left(\frac{s}{h}y\right)^2 = \frac{s^2}{h^2}y^2$.

$$V = \int_0^h \frac{s^2}{h^2}y^2 dy = \frac{s^2}{h^2} \cdot \frac{h^3}{3} = \frac{1}{3}s^2h$$

5.3 Volumes of Revolution: Disk Method

Rotating $y = f(x)$ around the x-axis from $x = a$ to $x = b$:

$$V = \pi \int_a^b [f(x)]^2 dx$$

Example: Rotate $y = \sqrt{x}$ from $x = 0$ to $x = 4$ around x-axis:

$$V = \pi \int_0^4 (\sqrt{x})^2 dx = \pi \int_0^4 x dx = \pi \left[\frac{x^2}{2} \right]_0^4 = \pi \cdot 8 = 8\pi$$

5.4 Volumes of Revolution: Washer Method

For regions between two curves $y = f(x)$ (top) and $y = g(x)$ (bottom), rotating around x-axis:

$$V = \pi \int_a^b ([f(x)]^2 - [g(x)]^2) dx$$

Example: Region between $y = x$ and $y = x^2$ from $x = 0$ to $x = 1$:

$$V = \pi \int_0^1 (x^2 - x^4) dx = \pi \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_0^1 = \pi \left(\frac{1}{3} - \frac{1}{5} \right) = \frac{2\pi}{15}$$

5.5 Volumes of Revolution: Shell Method

Rotating around y-axis, using cylindrical shells:

$$V = 2\pi \int_a^b x \cdot f(x) dx$$

Example: Rotate $y = x^2$ from $x = 0$ to $x = 2$ around y-axis:

$$V = 2\pi \int_0^2 x \cdot x^2 dx = 2\pi \int_0^2 x^3 dx = 2\pi \left[\frac{x^4}{4} \right]_0^2 = 2\pi \cdot 4 = 8\pi$$

5.6 Triple Integration for Volume

For a general 3D region D , volume is the integral of 1 over the region:

$$V = \iiint_D dV$$

5.7 Cylindrical Coordinates

For regions with cylindrical symmetry:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z$$

$$dV = r \, dr \, d\theta \, dz$$

$$V = \int_{\theta=0}^{2\pi} \int_{r=0}^{R(\theta)} \int_{z=z_1(r,\theta)}^{z_2(r,\theta)} r \, dz \, dr \, d\theta$$

Example: Volume of a cylinder of radius R and height h :

$$V = \int_0^{2\pi} \int_0^R \int_0^h r \, dz \, dr \, d\theta = \int_0^{2\pi} \int_0^R r h \, dr \, d\theta = \int_0^{2\pi} \frac{R^2 h}{2} d\theta = \pi R^2 h$$

5.8 Spherical Coordinates

For regions with spherical symmetry:

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi$$

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

$$V = \int_{\theta=0}^{2\pi} \int_{\phi=0}^{\pi} \int_{\rho=0}^{R(\theta,\phi)} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

Example: Volume of a sphere of radius R :

$$\begin{aligned}
V &= \int_0^{2\pi} \int_0^\pi \int_0^R \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \\
&= \int_0^{2\pi} \int_0^\pi \frac{R^3}{3} \sin \phi \, d\phi \, d\theta \\
&= \int_0^{2\pi} \frac{R^3}{3} [-\cos \phi]_0^\pi \, d\theta = \int_0^{2\pi} \frac{R^3}{3} (2) \, d\theta = \frac{4}{3} \pi R^3
\end{aligned}$$

6 MOMENT OF INERTIA FOR IRREGULAR SHAPES

6.1 General Formula

For any 3D object with density $\rho(\mathbf{r})$:

$$I = \iiint_V r_{\perp}^2 \rho(\mathbf{r}) dV$$

where $r_{\perp}$ is the perpendicular distance from the axis of rotation to the volume element dV .

6.2 Example: Irregular 2D Shape

For a flat, irregular plate rotating about an axis through its center:

1. Place shape in coordinate plane
2. Use double integration: $I = \iint_R (x^2 + y^2) \sigma dA$
3. If shape boundary is given by $y = f(x)$, set up:

$$I = \sigma \int_{x=a}^b \int_{y=f_1(x)}^{f_2(x)} (x^2 + y^2) dy dx$$

4. Evaluate inner integral first, then outer integral

7 MULTIPLE INTEGRATION

7.1 Double Integrals

Over a region R :

$$\iint_R f(x, y) dA$$

In Cartesian coordinates, $dA = dx dy$ or $dy dx$:

$$\iint_R f(x, y) dA = \int_{x=a}^b \int_{y=g_1(x)}^{g_2(x)} f(x, y) dy dx$$

Example: $\iint_R (x + y) \, dA$ where R is rectangle $0 \leq x \leq 2$, $0 \leq y \leq 1$:

$$\begin{aligned} \int_0^2 \int_0^1 (x + y) \, dy \, dx &= \int_0^2 \left[xy + \frac{y^2}{2} \right]_{y=0}^1 dx \\ &= \int_0^2 \left(x + \frac{1}{2} \right) dx = \left[\frac{x^2}{2} + \frac{x}{2} \right]_0^2 = 2 + 1 = 3 \end{aligned}$$

7.2 Double Integrals in Polar Coordinates

For regions with circular symmetry:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad dA = r \, dr \, d\theta$$

$$\iint_R f(x, y) \, dA = \int_{\theta=\alpha}^{\beta} \int_{r=r_1(\theta)}^{r_2(\theta)} f(r \cos \theta, r \sin \theta) r \, dr \, d\theta$$

Example: Area of a circle of radius R :

$$A = \iint_R dA = \int_0^{2\pi} \int_0^R r \, dr \, d\theta = \int_0^{2\pi} \frac{R^2}{2} \, d\theta = \pi R^2$$

7.3 Triple Integrals

Over a 3D region D :

$$\iiint_D f(x, y, z) \, dV$$

In Cartesian coordinates, $dV = dx \, dy \, dz$:

$$\iiint_D f(x, y, z) \, dV = \int_{x=a}^b \int_{y=g_1(x)}^{g_2(x)} \int_{z=h_1(x,y)}^{h_2(x,y)} f(x, y, z) \, dz \, dy \, dx$$

Example: Mass of a cube of side L with density $\rho(x, y, z) = xyz$:

$$\begin{aligned} M &= \int_0^L \int_0^L \int_0^L xyz \, dz \, dy \, dx \\ &= \int_0^L \int_0^L xy \left[\frac{z^2}{2} \right]_0^L \, dy \, dx = \int_0^L \int_0^L xy \frac{L^2}{2} \, dy \, dx \\ &= \frac{L^2}{2} \int_0^L x \left[\frac{y^2}{2} \right]_0^L \, dx = \frac{L^2}{2} \int_0^L x \frac{L^2}{2} \, dx \\ &= \frac{L^4}{4} \int_0^L x \, dx = \frac{L^4}{4} \cdot \frac{L^2}{2} = \frac{L^6}{8} \end{aligned}$$

7.4 Cylindrical Coordinates

For regions with cylindrical symmetry:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z$$

$$dV = r \, dr \, d\theta \, dz$$

$$\iiint_D f(x, y, z) \, dV = \int_{\theta=0}^{2\pi} \int_{r=0}^{R(\theta)} \int_{z=z_1(r,\theta)}^{z_2(r,\theta)} f(r, \theta, z) \, r \, dz \, dr \, d\theta$$

7.5 Spherical Coordinates

For regions with spherical symmetry:

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi$$

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

$$\iiint_D f(x, y, z) \, dV = \int_{\theta=0}^{2\pi} \int_{\phi=0}^{\pi} \int_{\rho=0}^{R(\theta,\phi)} f(\rho, \phi, \theta) \, \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

Integration Techniques Summary

Technique	When to Use
Power Rule	Polynomials, x^n
U-Substitution	Composition $f(g(x))g'(x)$
Integration by Parts	Products of different function types
Trig Substitution	$\sqrt{a^2 \pm x^2}$, $\sqrt{x^2 \pm a^2}$
Partial Fractions	Rational functions $\frac{P(x)}{Q(x)}$
Numerical Integration	No elementary antiderivative

Key Physics Formulas

$$\text{Mass: } M = \int \lambda(x) dx \quad (1\text{D}), \quad M = \iint \sigma(x, y) dA \quad (2\text{D}), \quad M = \iiint \rho(x, y, z) dV \quad (3\text{D})$$

$$\text{Center of Mass: } x_{cm} = \frac{1}{M} \int x dm, \quad y_{cm} = \frac{1}{M} \int y dm, \quad z_{cm} = \frac{1}{M} \int z dm$$

$$\text{Moment of Inertia: } I = \int r_{\perp}^2 dm$$

$$\text{Parallel Axis Theorem: } I = I_{cm} + Md^2$$

$$\text{Work: } W = \int \mathbf{F} \cdot d\mathbf{r}$$

$$\text{Hydrostatic Force: } F = \int \rho gh dA$$

$$\text{Volume: } V = \int A(x) dx \quad (\text{cross-sections}), \quad V = \iiint dV$$

Common Integrals in Physics

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C$$

$$\int \sin(ax) dx = -\frac{1}{a} \cos(ax) + C$$

$$\int \cos(ax) dx = \frac{1}{a} \sin(ax) + C$$

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin\left(\frac{x}{a}\right) + C$$

$$\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$$